



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 8 • STANDARD 6.DS.4

Mean, Median, and Mode

Lesson 8-2 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can find the mean, median, and mode of a data set.

Language Objective: I can explain my reasoning using the words mean, median, mode, and outlier.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Mean

Median

Mode

Outlier

Data distribution

Variability



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The average found by adding all values and dividing by how many there are is the

2

The middle value of a data set when the numbers are in order is the

3

The value that appears most often in a data set is the

4

A value that is much larger or smaller than the rest of the data is an

5

The way data values are spread out is the

6

How spread out the data values are is the

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

COMPARE

After ordering the scores on the number line, how do you find the median, and how is it different from the mean you calculated?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Mean** — The average. Add all the numbers, then divide by how many there are.
Media — El promedio. Suma todos los números y divide entre cuántos hay.

☐ **Median** — The middle number when you put them in order.
Mediana — El número del medio cuando los pones en orden.

☐ **Mode** — The number that shows up most often.
Moda — El número que aparece más veces.

☐ **Outlier** — A number that is much bigger or smaller than the rest.
Valor atípico — Un número mucho mayor o menor que los demás.

☐ **Data distribution** — How the data looks: where it sits and how spread out it is.
Distribución de datos — Cómo se ven los datos: dónde están y qué tan separados están.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

To find the median, I first ____ the scores, then pick the ____.

Para hallar la mediana, primero ____ los puntajes y luego elijo el ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Mean, median, and mode each describe the center of the data in a different way.

Evidence: Students distinguish the three measures and justify a choice, noting the mode (38) is the most repeated and the mean/median sit near the center.

Reasoning: This evidence connects the definition of mean to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **To find the median, I first ____ the scores, then pick the ____.**

Para hallar la mediana, primero ____ los puntajes y luego elijo el ____.

- **The mean is different because I have to ____ and ____.**

La media es diferente porque tengo que ____ y ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Mean
2. **Notes 2:** Median
3. **Notes 3:** Mode
4. **Notes 4:** Outlier
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Mean, Median, and Mode is finding the median without ordering the data first — students pick the middle number straight from the list as given (for 9, 4, 7, 2, 6 they say the median is 7, the middle position in that original order) instead of sorting least to greatest (2, 4, 6, 7, 9) and finding the true middle value, 6. Before finalizing a median, always reorder the data smallest to largest, then count to the middle position — and if there is an even number of values, average the two middle numbers.*
8. **Try It 1:** A. 4 — *The mode is the value that appears most often. 4 shows up twice; every other value appears once. So the mode is 4.*
9. **Try It 2:** A. The median, because the outlier barely moves the middle value — *A high outlier pulls the mean upward, making it larger than most games. The median is the middle of the ordered data, so one unusual game barely changes it — it best describes a typical game.*